

* Assignment No :- 04

(Q.1)
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Feature	Unicast	Multicast	Broadcast
Definition	A communication where a message is sent from one sender to one receiver.	A communication where a message is sent from one sender to group of receiver.	A communication where a message is sent from one sender to all receivers.
Transmission	In unicast, data is sent to a single recipient.	In multicast, data is sent to a group of recipients.	In Broadcast, data is sent to all recipients in a network.
Addressing	Unicast uses a unique destination address.	Multicast uses a special multicast address.	Broadcast uses a special broadcast address.
Delivery	Guaranteed delivery of a data to receiver.	Not all devices may be interested in the data.	Not all devices may be interested in the data.
Network Traffic	Generates the least amount of network traffic.	Generates moderate network traffic.	Generates the most amount of network traffic.

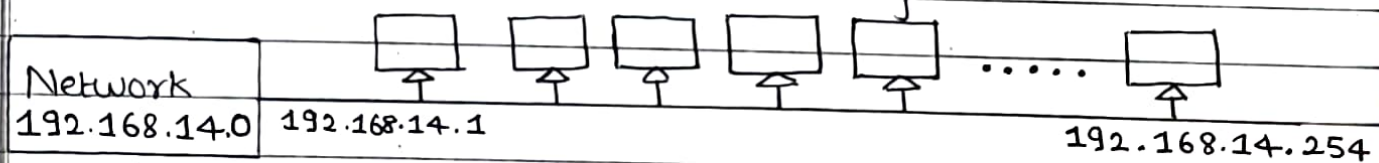
	Unicast	Multicast	Broadcast
Security	More secure because data is sent to a specific recipient.	Moderately secure because data is sent to a specific group of devices.	Less secure because data is sent to all devices in the network.
Examples	Email, File transfer	Video streaming, online gaming	DHCP requests, ARP requests
Destination	The destination is only one single receiver.	The destination is selected group of receivers.	The destination includes all devices in the network.
Bandwidth Usage	Moderate: only one copy of the data is sent, so it doesn't overload the network.	Moderate: only selected devices get the data, so it avoids the wastage of broadcast.	High: Every device gets a copy of data, even if most are not interested, which wastes bandwidth.
Latency	Low: Direct one-to-one delivery makes it fast, with no extra processing.	Moderate: Delivery to a group requires extra routing decisions, so slightly slower than unicast but faster than broadcast.	High: Network becomes congested as all devices process the data, causing delays.

Feature	classful Addressing	classless Addressing
Division	IP addresses divided into fixed classes (A, B, C, D, E).	No fixed classes, uses variable-length subnet masks.
Subnet Mask	Default Subnet mask fixed for each class (e.g., class C $\rightarrow$ 255.255.255.0)	Subnet mask defined by prefix length (e.g., /27, /29).
Address Utilization	Wastage of IP addresses due to fixed block sizes.	Efficient allocation, no wastage of addresses.
Network & Host Flexibility	The Network ID and Host ID are adjusted according to the classes.	Network ID and Host ID distinction does not exist with classless addressing.
Flexibility	Limited by predefined classes.	Highly flexible, supports custom Subnetting.
Number of Networks	Limited number of networks due to fixed classes.	Large number of networks possible, because of flexible subnetting.
Scalability	Not Scalable for modern internet growth.	Scalable, supports VLSM (Variable Length Subnet masking).
Example:-	class C $\rightarrow$ 192.168.1.0 (256 addresses even if only 50 are needed).	192.168.1.0/27 $\rightarrow$ (only 32 addresses allocated, saving space).

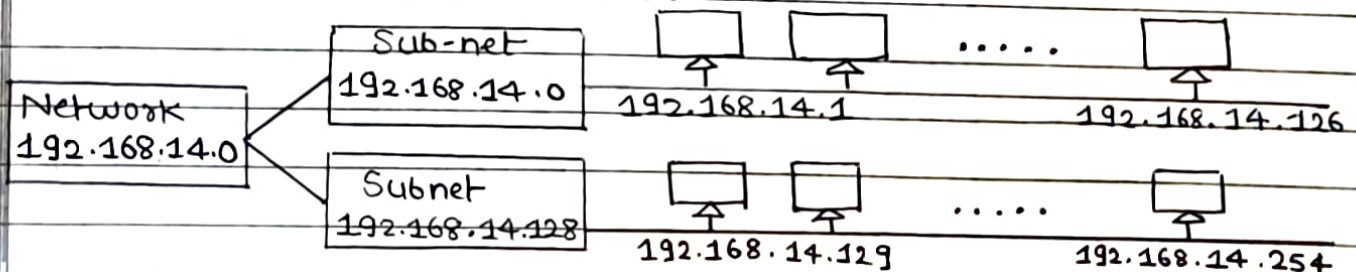
• Subnetting of IPv4 :-

- Subnetting is a method of dividing a single physical network into numerous smaller logical sub-networks.
- These subnetworks are referred to as subnets. An IP address is formed by combining a network and host segments.
- A subnet is created by accepting bits from the IP address host part and is used to split the original network into smaller subnetworks.
- The process of subnetting involves turning host bits into network bits.
- It permits the administrator to split single class A, class B or class C network into smaller sections.
- VLSM (Variable Length Subnet Mask) divides IP address space into subnets of varying sizes into smaller sections.

• A class C network without Sub-netting :-



• A class C network with two subnets :-



- Classes of IPv4 addresses :-

Address Class	1 st Octet range in decimal	1 st Octet Bits	Network (N) and Host (H) partition	Default Mask (Decimal)	No. of possible networks and hosts per network.
A	0-127	00000000 - 01111111	N.H.H.H	255.0.0.0	128 Nets (2^7) 16,777,214 hosts ($2^{24} - 2$)
B	128 - 191	10000000 - 10111111	N.N.H.H	255.255.0.0	16,384 Nets (2^{14}) 65,534 hosts ($2^{16} - 2$)
C	192 - 223	11000000 - 11011111	N.N.N.H	255.255.255.0	2,097,152 Nets (2^{21}) 254 hosts ($2^8 - 2$)
D	224 - 239	11100000 - 11101111	NA (Multicast)	-	-
E	240 - 255	11110000 - 11111111	NA (Experimental)	-	-

- Subnetting - 5 steps

1. Identify class of IP address and note default subnet mask.
2. Convert default subnet mask into Binary.
3. Note the no. of hosts required per subnet and find the Subnet Generator (SG) and octet position.

4. Generate the new subnet mask.
5. Use the SG and generate the network ranges (Subnets) in the appropriate octet position.

• example :-

Subnet the IP address 216.21.5.0^{into 30} hosts in each subnet.

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1. class C - Default Subnet mask : 255.255.255.0

2. 11111111.11111111.11111111.00000000

3. No. of hosts / Subnet : 30 (11110) - 5 bits

SG = 32, Octet position = 4

11111111.11111111.11111111.11100000

4. New Subnet mask : 255.255.255.224 or /27

5. Network Ranges (Subnets):-

216.21.5.0 - 216.21.5.31

216.21.5.32 - 216.21.5.63

216.21.5.64 - 216.21.5.95

216.21.5.96 - 216.21.5.127

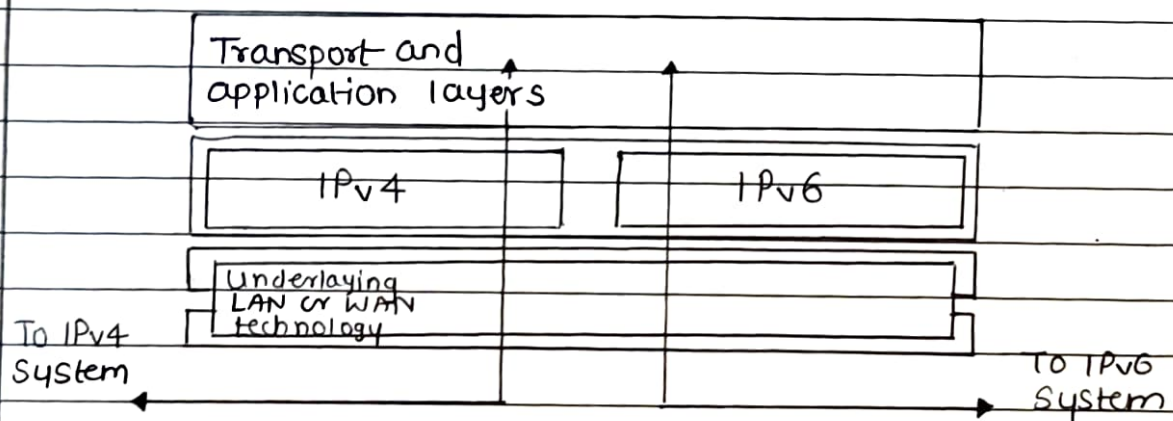
216.21.5.128 - 216.21.5.159

Transition from IPv4 to IPv6 :-

- The transition from IPv4 to IPv6 is happening because the IPv4 address space (32-bit) has been nearly exhausted, while IPv6 provides a much larger pool of addresses.
- Because of the huge number of systems on the internet, the transition from IPv4 to IPv6 cannot happen suddenly.
- It takes a considerable amount of time before every system in the internet can move from IPv4 to IPv6.
- There are three strategies for ~~transmis~~ transition from IPv4 to IPv6.

1. Dual Stack :-

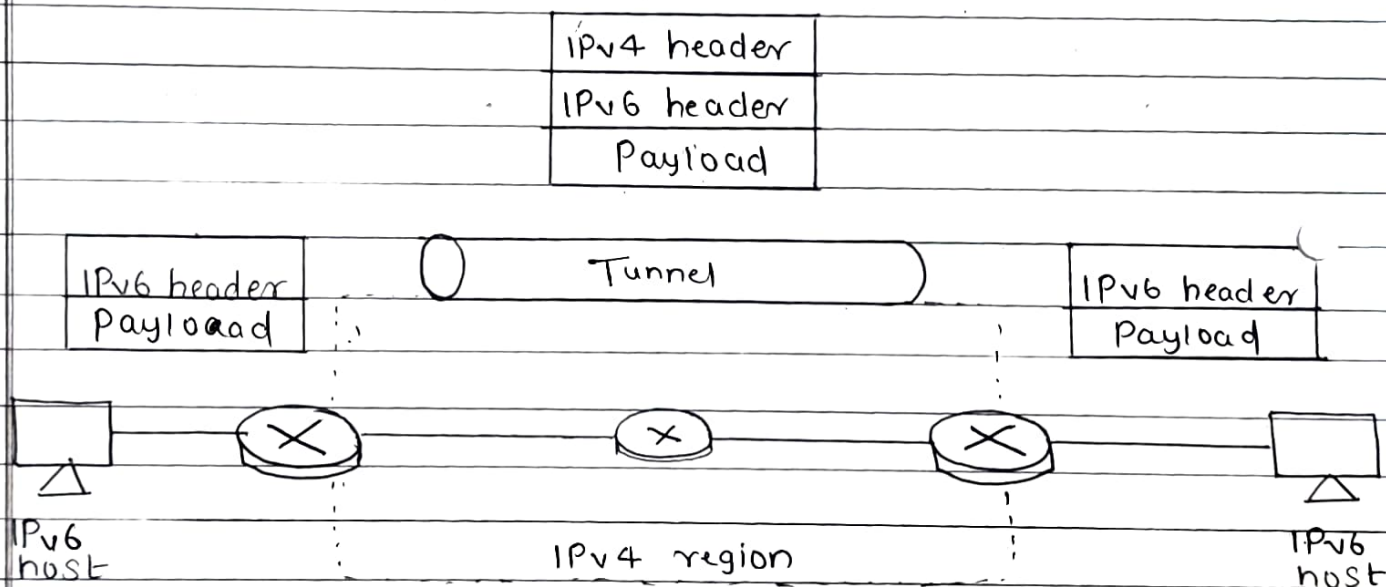
- Dual Stack allows devices to run both IPv4 and IPv6 at the same time, choosing the protocol supported by the other end. If IPv6 is available, it is used; otherwise, communication falls back to IPv4. It is widely supported by ISPs, operating systems, and modern networks.



- To determine which version to use when sending packet to destination, source host queries DNS. If DNS returns IPv4 address, source host send IPv4 packet. If DNS returns IPv6 address, source host send IPv6 packet.

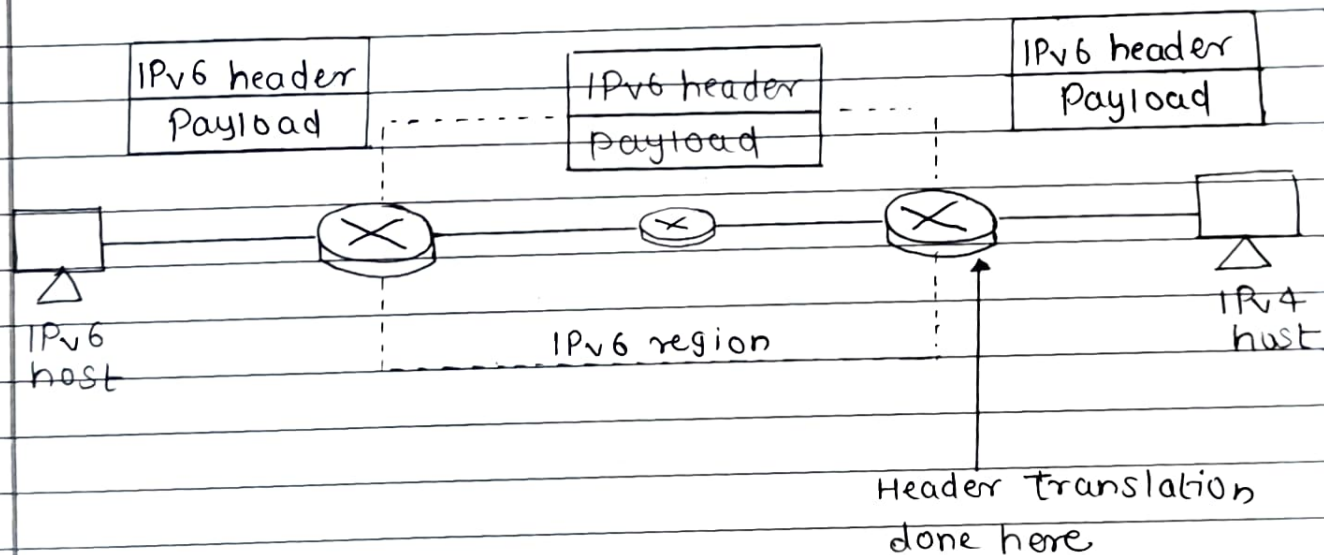
2. Tunneling :-

- Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses IPv4.
- To pass through the region, the packet must have an IPv4 address.
- So the IPv6 packet is encapsulated in an IPv4 packet when it enters the region, and it leaves its capsule when it exits the region.
- It seems as if the IPv6 packet goes through a tunnel at one end emerges at the other end.
- To make it clear that the IPv4 packet is carrying an IPv6 packet as data, the protocol value is set to 41.



3. Header Translation :-

- Header translation is necessary when the majority of the internet has moved to IPv6 but some systems still use IPv4.
- The sender wants to use IPv6, but the receiver does not understand IPv6.
- Tunneling does not work in this situation, because the packet must be in the IPv4 format to be understood by the receiver.
- In this case, the header format must be totally changed through header translation.



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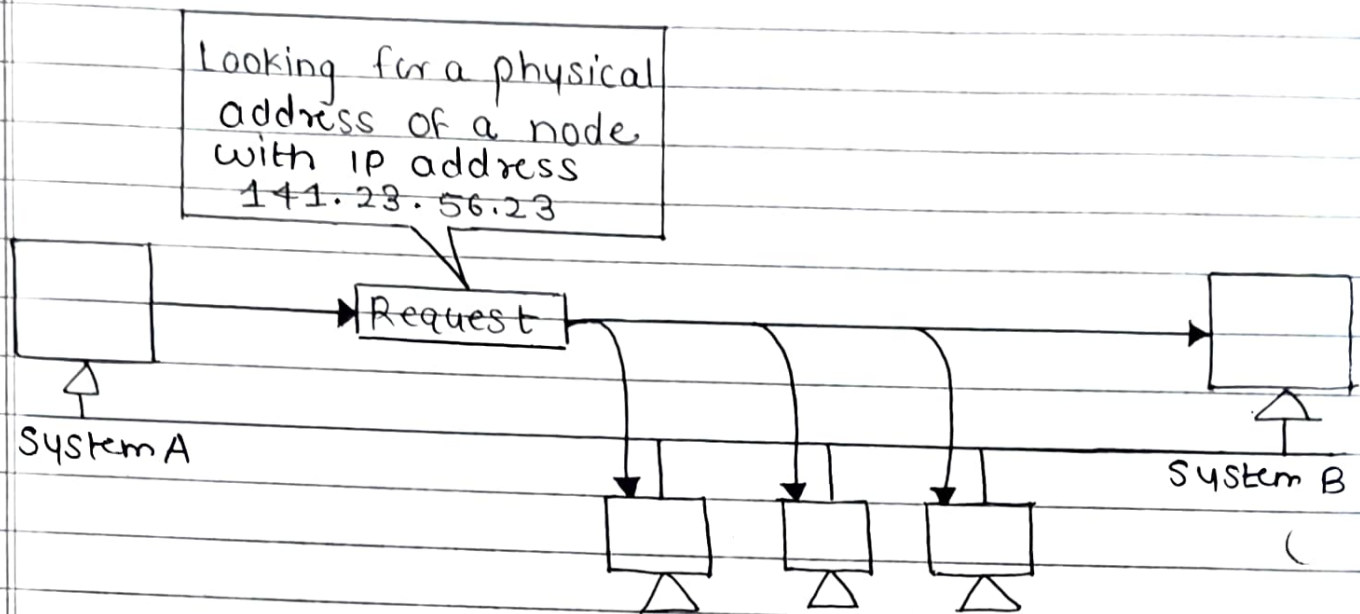


Feature	Dijkstra's Shortest path	Link State Routing	Distance Vector Routing
Definition	Algorithm to compute shortest path from one node to all others.	Routing protocol where routers know the full network topology and compute shortest paths using Dijkstra's algorithm.	Routing protocol where routers exchange distance (cost) information with neighbors until the best path is found.
Routing information	Needs full network graph as input.	Each router collects link states from all routers to build global topology.	Each router only knows the distance vectors received from neighbors.
Algorithm used	Dijkstra's Shortest path uses Dijkstra (greedy) algorithm.	Link state routing protocol uses Dijkstra (per router) algorithm.	Distance Vector Routing protocol uses Bellman Ford algorithm.
Scalability	Good for small to medium networks.	Link State Routing protocol is good for large networks.	Distance Vector Routing protocol is poor for large networks.
Examples Usage	Used in graph theory and as basis for routing.	OSPF, IS-IS (modern protocols).	RIP, IGRP (older / simple protocols).

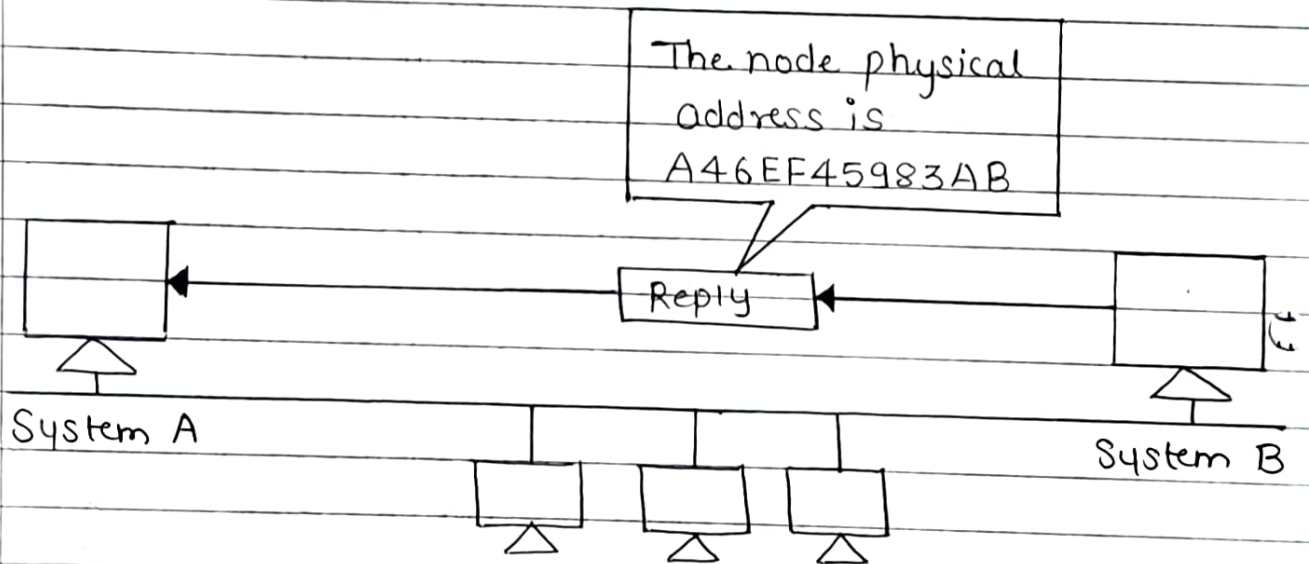
• ARP protocol :-

- ARP stands for Address Resolution Protocol.
- ARP is used to convert logical address i.e., IP address into physical address i.e., MAC address.
- For communication, the sender needs the MAC address of the destination node.
- When a node wants the MAC address of another node in same network, it sends an ARP query packet.
- This query contains the source IP & MAC address, and only the IP address of the destination.
- The query is broadcast to all nodes in the network.
- The node with the matching IP address replies with its MAC address.
- Frequent ARP requests increase network traffic.
- To reduce this, system maintain an ARP cache with recently resolved IP-MAC bindings.

• working: The host broadcasts an ARP inquiry packet containing the IP address over the network in order to find out the physical address of another computer on its network. The ARP packet is received and processed by all hosts on the network; however only the intended recipient can identify the IP address and reply with physical address. After adding the physical address to the datagram header and cache memory, the host storing the datagram transmits it back to the sender.



ARP request is broadcast

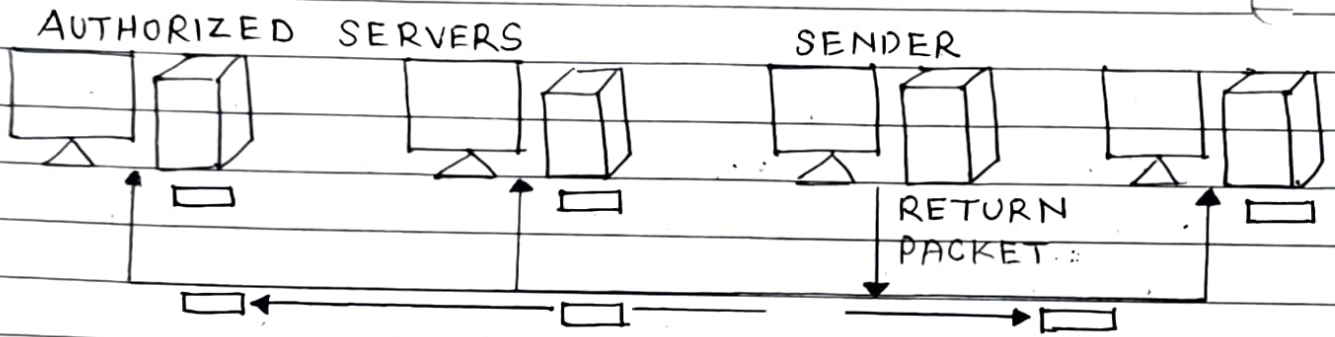
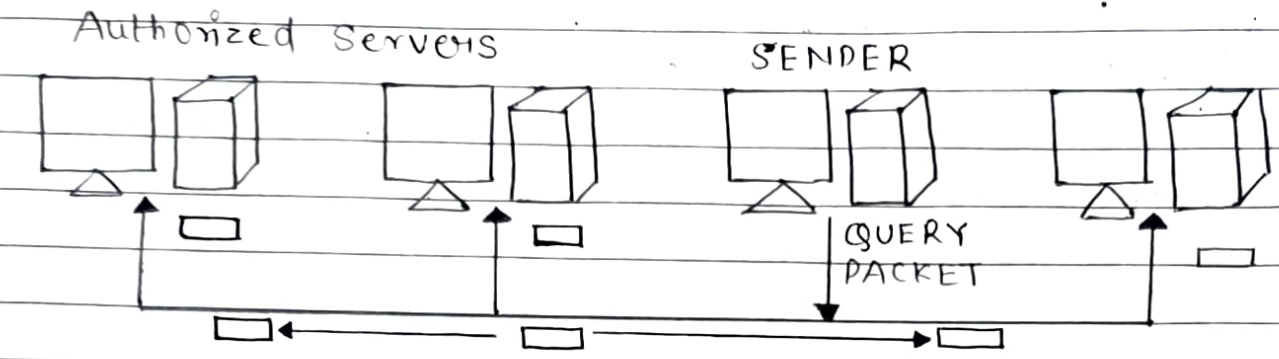


ARP reply is unicast

• RARP protocol :-

- RARP stands for Reverse Address Resolution Protocol.
- RARP works opposite of ARP.
- RARP is used to convert MAC address into IP address (i.e., physical address into logical address).
- RARP provides with a feature for the systems and applications to get their own IP address from a DNS (Domain Name System) or router.
- This type of resolution is required for various tasks such as executing reverse DNS lookup.
- As reverse address resolution protocol works at low level it requires direct network addresses.
- The reply from server mostly carries a small information but the 32 bit internet address is used and it does not exploit the full potential of a network such as ethernet.

• Working :- RARP works at the Network Access Layer to send data between two places. Each device has two addresses: MAC (physical) and IP (logical). IP address is assigned by software; MAC address is built into hardware. A normal computer can act as a RARP server if it stores MAC-IP mappings. Only RARP servers can reply to client RARP requests. Requests are sent using physical address and Ethernet broadcast. The server responds by providing the client its IP address.



◦ ICMP protocol :-

- ICMP stands for Internet Control Message Protocol.
- ICMP is a part of IP protocol suite.
- ICMP is an error reporting and network diagnostic protocol.
- Feedback in the network is reported to designated host. Meanwhile, if any kind of error occurs it is then reported to ICMP.
- ICMP protocol consist of many error reporting and diagnostic messages.
- ICMP protocol handles various kinds of errors such as time exceeded, redirection, source quench, destination unreachable,

parameter problems etc..

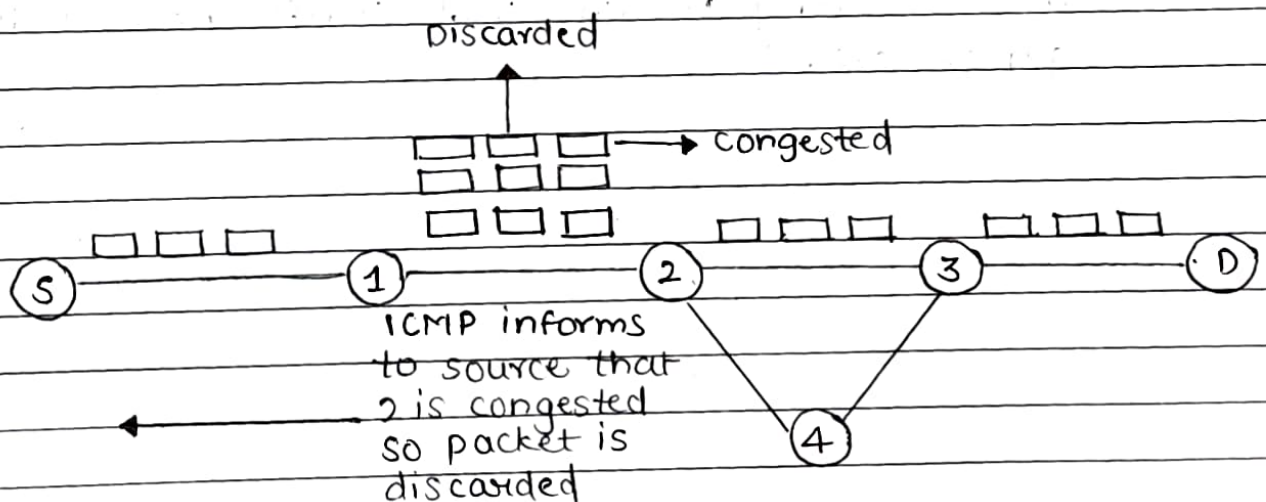
The messages in ICMP are divided into two types :-

- Feedback in the network is reported Error Message :

Error message, States about the issues or problems that are faced by the host or routers during processing of IP packet..

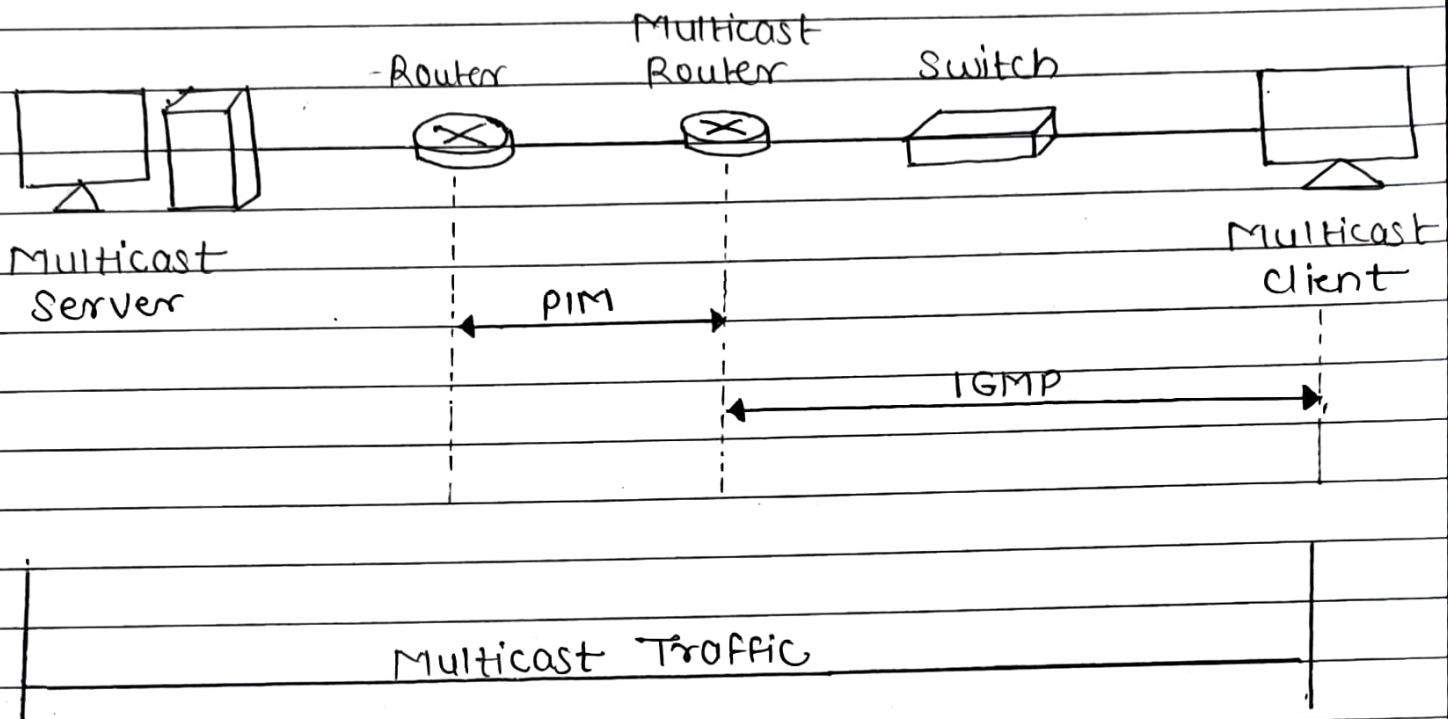
- Query Message : Query messages are used by the host in order to get information from a router or another host.

- working :- Unlike TCP and UDP, ICMP is a connectionless protocol; meaning it doesn't require a connection to be established with the target device, in order to transmit a message. TCP and ICMP operate differently from one another, TCP is connection-oriented protocol while ICMP operates without a connection. Every time a connection is made prior to a message being sent, a TCP handshake is required of both devices. Datagrams including an IP header containing ICMP data are used to transmit ICMP packets.



IGMP protocol :-

- IGMP stands for Internet Group Message Protocol. It is a multicasting communication protocol.
- It utilizes the resources efficiently while broadcasting the messages and data packets. IGMP is also a protocol used by TCP/IP.
- Other hosts connected in the network and routers makes use of IGMP for multicasting communication that have IP networks.
- In many networks multicast routers are used in order to transmit the messages to all the nodes.
- Multicast routers therefore receives large number of packets that needs to be sent. But to broadcast this packet is difficult as it would increase the overall network load.
- Therefore IGMP helps the multicast routers by addressing them while broadcasting.
- As multicast communication consists of more than one senders and receivers, the IGMP is majorly used in various applications such as streaming-media, web conference tools, games, etc.



(Internet Group Message Protocol)